AGRIVIVE / BRAHMA-ROOT

ASSAM / RAISED-BED CONTEXT / PASSIVE ROOT-ZONE AIR CHIMNEY

SERIES 03 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



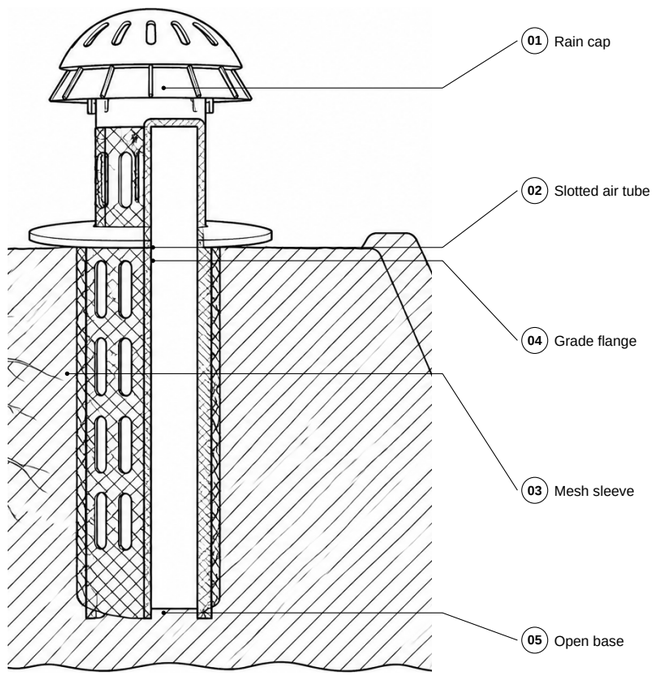
STATE CONTEXT / INDIGENOUS INSPIRATION

Raised and sunken bed systems are studied in Assam for seasonally waterlogged agricultural land. BRAHMA-ROOT proposes a new screened air chimney in a raised bed; it does not establish a drainage or crop-yield improvement.

Practice source: AAU integrated farming systems research - raised and sunken beds | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 60 tube OD x 240 overall H

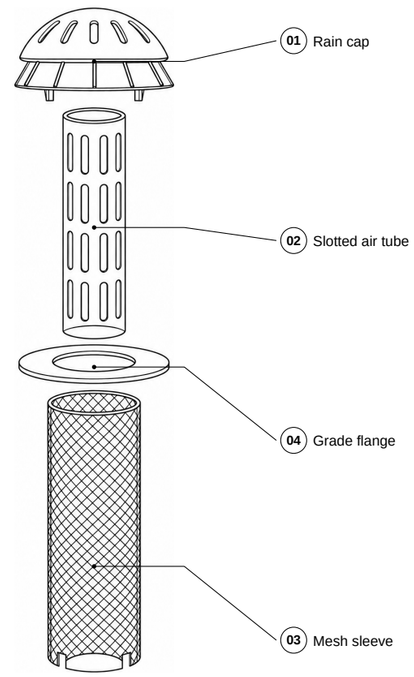


DESIGN / OPERATION NOTES

Seat the screened tube in a trial raised bed with grade flange at the soil surface.
Keep the protected upper air openings and lower passage unobstructed.
Remove the cap for inspection; the hollow bore is an air space, not a water reservoir.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
60 tube OD x 240 overall H

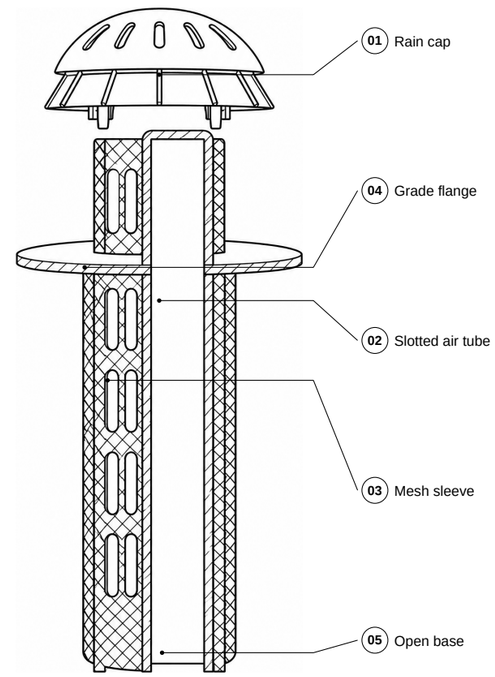


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

SECTION BASIS / mm
150 buried + 90 above grade = 240 overall H; 52 bore ID; flange/cap diameters TBD



DESIGN / OPERATION NOTES

Keep the protected upper air openings and lower passage unobstructed.
Remove the cap for inspection; the hollow bore is an air space, not a water reservoir.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Tube OD / overall H	**60 / 240**
Clear bore ID	**52**
Buried / above-grade H	**150 / 90**

02 GEOMETRIC CHECK

BORE CROSS-SECTION

2,123.7 mm2

pi/4 x 52^2
Air space; no storage or drainage capacity claim

03 PARTS / QTY

01 Rain cap	**1**
02 Slotted air tube	**1**
03 Mesh sleeve	**1**
04 Grade flange	**1**
05 Open base	**Integral**

Tube/cap/flange: candidate polymer. Soil screen: candidate mesh; aperture TBD.

04 INTERFACES / CHECKS

Test mesh clogging and soil entry.

Measure root-zone conditions before inferring benefit.

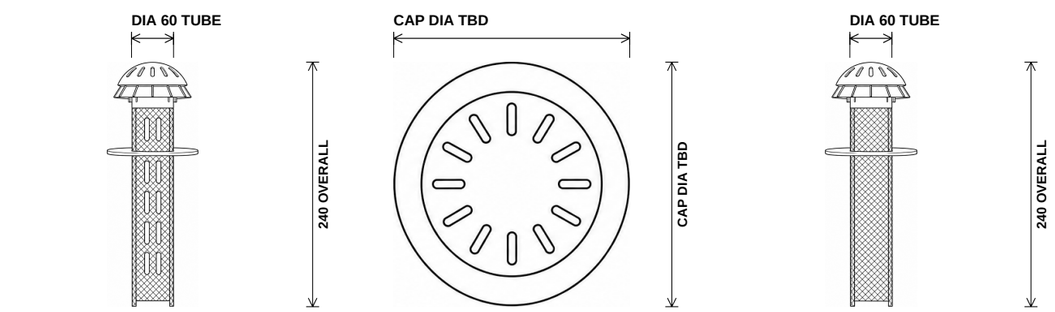
Check cap retention and inspection access.

05 DATUMS / DRAWING BASIS

A: tube axis. B: underside of grade flange; 150 below B and 90 above B.

PRELIMINARY / VALIDATION PENDING

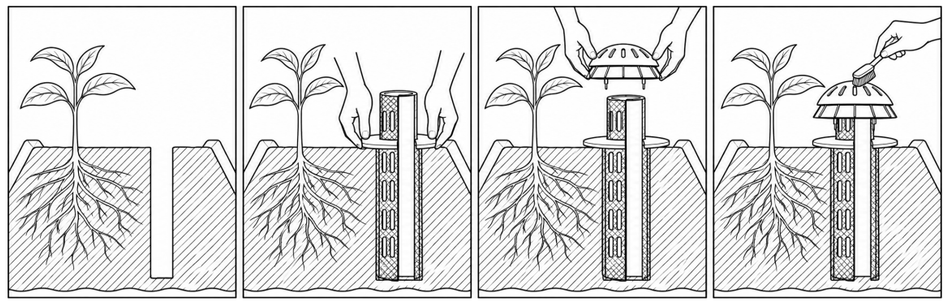
03 ORTHOGRAPHIC STUDIES / mm



FRONT / TUBE **TOP / CAP** **SIDE / TUBE**

Tube DIA 60 OD / DIA 52 ID. Height chain: 150 buried + 90 above grade = 240 overall. Cap / flange diameters TBD.

05 INSTALLATION / OPERATING SEQUENCE



1 PREPARE HOLE **2 SET TUBE** **3 FIT CAP** **4 CLEAN / INSPECT**

Concept sequence; validate fit, access and operating conditions in prototypes.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-03 / P01	BRAHMA-ROOT	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	03 / 28